

CSE331 - Automata and Computability

Quiz 5

Section 16 (Set A)

Full name:

ID:

Total Marks: 20

1. Draw PDA of the following languages:

- a. $w_1\#w_2$, w_1 consists of an odd number of 0s only, w_2 contains 011 as a substring.
- b. $0^i1^j2^k$, $i, j, k \geq 0$, $j > 2i + k$
- c. w_1w_2 , $|w_2|=2|w_1|$, there are odd number of 1s in w_1 . [alternate: w_1w_2 , $|w_2|=2|w_1|$, there are odd number of 1s in w_2]

2. Bonus: Write whether the following statements are correct. If not, explain why it is incorrect:
- a. Context-free languages are closed under union, complement and intersection operations.
 - b. From the derivation $B \rightarrow aBb \rightarrow aab$, aab is a valid string of the CFG:
$$A \rightarrow aB$$
$$B \rightarrow aBb \mid a$$
 - c. Regular languages are a subset of CFG.

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Quiz 5

Section 16 (Set B)

Full name:

ID:

Total Marks: 20

1. Draw PDA of the following languages:

- a. $w_1\#w_2$, w_1 consists of an even number of 0s only, w_2 contains 101 as a substring.
- b. $0^i1^j2^k$, $i, j, k \geq 0$, $j > i + 3k$
- c. w_1w_2 , $|w_2|=2|w_1|$, there are even number of 1s in w_2

2. Bonus: Write whether the following statements are correct. If not, explain why it is incorrect:
- a. Context-free languages are closed under union, complement and intersection operations.
 - b. From the derivation $B \rightarrow aBb \rightarrow aab$, aab is a valid string of the CFG:
$$A \rightarrow aB$$
$$B \rightarrow aBb \mid a$$
 - c. Regular languages are a subset of CFG.